Hackathon 2026 :D

Corn Yield Predictor Project

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CSU Hackathon 2026 · USDA NASS Challenge

The Problem

- Due to decreasing number of returned surveys, the ability to predict corn crop yield has dwindled. With satellite imagery and weather data, we can predict the expected corn yield in bushels per acre.
- Currently with our model, we have issues with volitility in the models prediction. We drastically undershot the predictions for Iowa, Wisconsin, Nebraska, Missouri and we overshot the Colorado prediction barely.
- Given the model has more time to train and more data to train, we are confident that this model will accurately predict the bushels of corn per acre in the United States

> *"Maybe its a shortcut, Dwight. It said to go right...*
>
> *The machine knows. Stop yelling at me"*
>
> — The Office

Methodology

1

Gather Sources

Use sources provided from NASA and USDA to collect information.

2

Parse Files

Sort through data to narrow information.

3

Prediction Modelling

Use data to make a model to predict future crop yields.

We tried.

Data Pipeline

Cropland Data layer

Contains raster data that allows us to get the cover of corn

NASS Quickstats CSVs

Forecasted values by months, for corn yield in bushel per acre

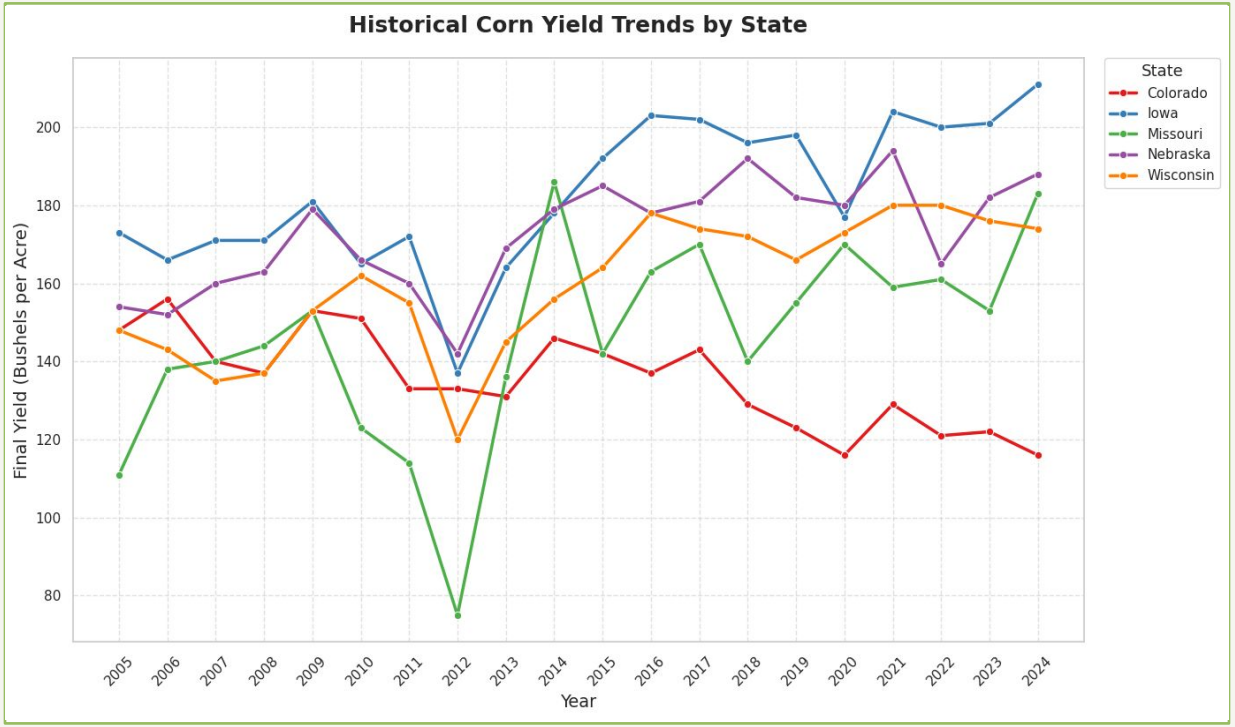
NASA Hls Sattelite Images

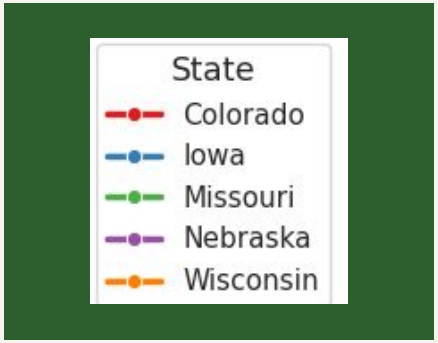
Mask with cdl to filter for state data to aggregate ndvi

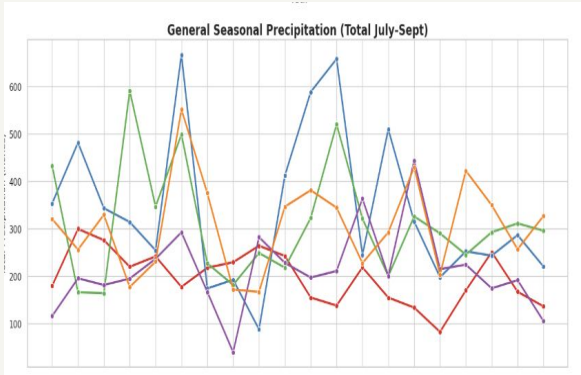
NASA Daymet

Access weather data for our cone of uncertainty algorithm

Results from Data Piplines

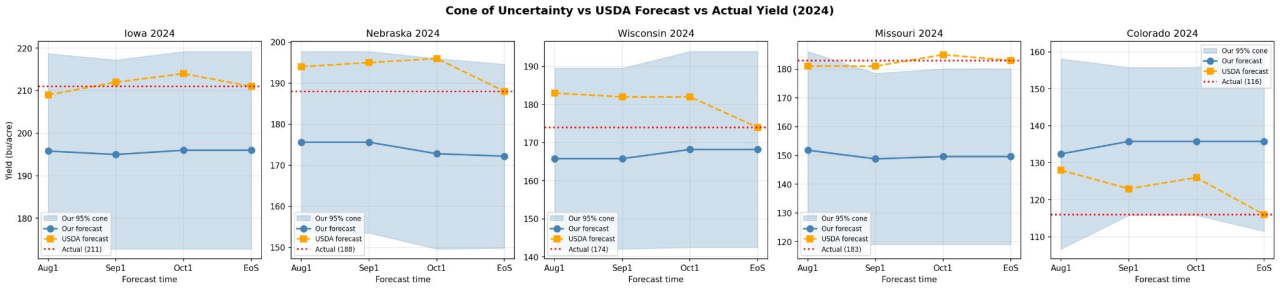






Pattern of Corn Yields

Cone of Uncertainty

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Total forecasts: 60

=== Coverage ===
Overall: 51/60 = 85.0%

=== Coverage by Year ===
      sum  count    pct
Year
2022   14     20   70.0
2023   20     20  100.0
2024   17     20   85.0

=== Errors (vs Final Yield) ===
Your MAE:  15.4 bu/acre
Your RMSE: 17.3 bu/acre
USDA MAE:  4.7 bu/acre
USDA RMSE: 6.3 bu/acre
```

- **60 independent forecasts across 3 hold-out years**
- **100% coverage in typical year (2023) demonstrates baseline reliability**
- **70% in 2022 drought reveals limitation: analog methods need historical analogs**

85%

Cone coverage 17/20

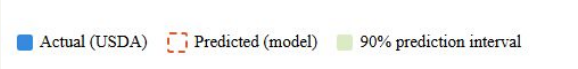
17.6

Our MAE bu/acre

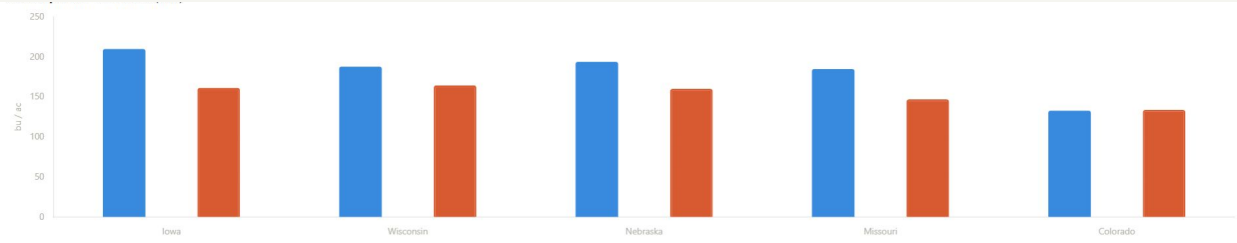
4.3

USDA MAE bu/acre

Predicted/Exact

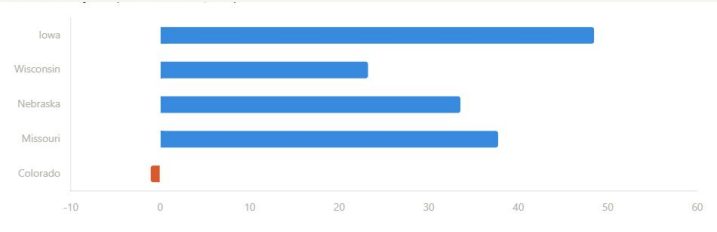


Actual vs predicted — final forecast (bu/ac)



This model kept the predictions conservative and consistently undershot the exact bu/acre.

Forecast error by state (Actual − Predicted, bu/ac)



We had pretty consistent forecasted error rates except for Colorado where we had our most accurate prediction, with it overshooting by 1.06 bu/acre.

Prediction interval vs actual — final forecast

The green band is our 90% confidence window — meaning we expected the actual to land inside it 90% of the time. It did for Colorado, but missed for all four others. We are unsure why data got skewed so much.

Thank You

Questions?

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